OUTPUT

Testing:

```
PS C:\Users\Owner\Documents\food_donation_system\food_donation_system>
python -m unittest -v
test_failures (tests.test_workflow.LocationServiceTests.test_failures) ... ok
test_success_falls_back_to_city
(tests.test_workflow.LocationServiceTests.test_success_falls_back_to_city)
... ok
test_available_until
(tests.test_workflow.ValidationTests.test_available_until) ... ok
test_capitalisation (tests.test_workflow.ValidationTests.test_capitalisation)
... ok
test_phone (tests.test_workflow.ValidationTests.test_phone) ... ok
test_quantity (tests.test_workflow.ValidationTests.test_quantity) ... ok
test_cancel_then_claim_rejected
(tests.test_workflow.WorkflowTests.test_cancel_then_claim_rejected) ... ok
test_corrupt_file_is_backed_up
(tests.test_workflow.WorkflowTests.test_corrupt_file_is_backed_up) ... ok
test_double_claim_rejected
(tests.test_workflow.WorkflowTests.test_double_claim_rejected) ... ok
test_duplicate_phone_rejected
(tests.test_workflow.WorkflowTests.test_duplicate_phone_rejected) ... ok
test_expired_donations_leave_available_list
(tests.test_workflow.WorkflowTests.test_expired_donations_leave_available_lis
t) ... ok
test_failed_save_rolls_back
(tests.test_workflow.WorkflowTests.test_failed_save_rolls_back) ... ok
test_full_lifecycle (tests.test_workflow.WorkflowTests.test_full_lifecycle)
... ok
test_ids_follow_expected_pattern
(tests.test_workflow.WorkflowTests.test_ids_follow_expected_pattern) ... ok
test_location_api_failures_are_not_fatal
(tests.test_workflow.WorkflowTests.test_location_api_failures_are_not_fatal)
... ok
test_only_assigned_volunteer_can_progress
(tests.test_workflow.WorkflowTests.test_only_assigned_volunteer_can_progress)
... ok
test_records_survive_restart
(tests.test_workflow.WorkflowTests.test_records_survive_restart) ... ok
test_release_makes_donation_available_again
(tests.test_workflow.WorkflowTests.test_release_makes_donation_available_agai
n) ... ok
test_roles_are_enforced
(tests.test_workflow.WorkflowTests.test_roles_are_enforced) ... ok
test_steps_cannot_be_skipped_or_reversed
(tests.test_workflow.WorkflowTests.test_steps_cannot_be_skipped_or_reversed)
... ok
```

```
test_unknown_ids (tests.test_workflow.WorkflowTests.test_unknown_ids) ... ok

----------------------------------------------------------------------
Ran 21 tests in 0.112s

OK
```

OUTPUT:

Registration:

```
PS C:\Users\Owner\Documents\food_donation_system\food_donation_system>
python main.py
Food Donation Management System (Prototype)

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 1

Register User
Role (Donor/Volunteer/Requester): Donor
Name: Paul
Phone: 9876543210
Location: chennai
Registered successfully. Your Donor ID is DNR002.

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 1

Register User
Role (Donor/Volunteer/Requester): Volunteer
Name: Andrews
Phone: 8907652341
Location: chennai
Registered successfully. Your Volunteer ID is VOL103.

Main Menu
1. Register User
```

```
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 1

Register User
Role (Donor/Volunteer/Requester): Requester
Name: Thasneem
Phone: 8135790246
Location: chennai
Registered successfully. Your Requester ID is REQ002.

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 2
Enter your User ID: REQ002
Welcome, Thasneem (Requester).

Requester Menu - REQ002 (Thasneem)
1. View Available Donations
2. Logout
Enter your choice (1-2): 1

Available Donations
-------------------
No donations are available right now.

Requester Menu - REQ002 (Thasneem)
1. View Available Donations
2. Logout
Enter your choice (1-2): 2
Logged out.

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 3
Enter Donation ID: DNR002
Error: No donation found with ID 'DNR002'.
```

```
Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 4

All Donations
No donations found.

Donation Summary
----------------
Available : 0
Claimed : 0
Picked Up : 0
Delivered : 0
Cancelled : 0
Expired : 0
Total : 0

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 5

Registered Users
ID      Role       Name          Phone        Location
-----------------------------------------------------
DNR001  Donor      Ravi Kumar    98765 43210  Chennai
VOL101  Volunteer  Priya         91234 56780  Chennai
VOL102  Volunteer  Arun          99887 76655  Chennai
REQ001  Requester  Hope Shelter  94455 66778  Chennai
DNR002  Donor      Paul          98765 43210  chennai
VOL103  Volunteer  Andrews       89076 52341  chennai
REQ002  Requester  Thasneem      81357 90246  chennai

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
```

```
Enter your choice (1-6): 6
All records are saved. Goodbye!
```

Working:

```
Food Donation Management System (Prototype)

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 2
Enter your User ID: DNR002
Welcome, Paul (Donor).

Donor Menu - DNR002 (Paul)
1. Create Donation
2. My Donations
3. Cancel Donation
4. Logout
Enter your choice (1-4): 1

Create Donation
Food Name: Biriyaani
Quantity: 20kg
Location: chennai
Available Until: 26-10-2026
  Enter a time like 8:00 PM or 20:00 (optionally 'tomorrow 9 AM' or '2026-10-
26 8:00 PM').
Available Until: 20:00
Contact Number [98765 43210]: y
  Enter a valid 10-digit mobile number, e.g. 9876543210.
Contact Number [98765 43210]: 9876543210

Donation Created Successfully.
Donation ID: DON001
Status: Available
Location verified: Chennai, Tamil Nadu, India

Donor Menu - DNR002 (Paul)
1. Create Donation
2. My Donations
3. Cancel Donation
4. Logout
Enter your choice (1-4): 2

My Donations
```

```
ID      Food       Quantity  Location  Until    Status     Volunteer
-------------------------------------------------------------------
DON001  Biriyaani  20 Kg     Chennai   8:00 PM  Available  -

Donor Menu - DNR002 (Paul)
1. Create Donation
2. My Donations
3. Cancel Donation
4. Logout
Enter your choice (1-4): 4
Logged out.

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 2
Enter your User ID: VOL103
Welcome, Andrews (Volunteer).

Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 1

Available Donations
-------------------
DON001
Food : Biriyaani
Quantity : 20 Kg
Location : Chennai
Status : Available

Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 2
```

```
Enter Donation ID to claim: DON001

Donation DON001
Status: Claimed
Volunteer: VOL103
Pickup contact: Paul, 98765 43210

Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 3
Enter Donation ID picked up: DON001

Donation DON001
Status: Picked Up
Volunteer: VOL103

Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 6

My Claimed Donations
ID      Food       Quantity  Location  Until    Status     Volunteer
-------------------------------------------------------------------
DON001  Biriyaani  20 Kg     Chennai   8:00 PM  Picked Up  VOL103

Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 7
Logged out.

Main Menu
1. Register User
```

```
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 2
Enter your User ID: REQ002
Welcome, Thasneem (Requester).

Requester Menu - REQ002 (Thasneem)
1. View Available Donations
2. Logout
Enter your choice (1-2): 1

Available Donations
-------------------
No donations are available right now.

Requester Menu - REQ002 (Thasneem)
1. View Available Donations
2. Logout
Enter your choice (1-2): 2
Logged out.

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 2
Enter your User ID: VOL103
Welcome, Andrews (Volunteer).

Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 4
Enter Donation ID delivered: DON001

Donation DON001
Status: Delivered
Volunteer: VOL103
```

```
Volunteer Menu - VOL103 (Andrews)
1. View Available Donations
2. Claim Donation
3. Mark Picked Up
4. Mark Delivered
5. Release Donation
6. My Claimed Donations
7. Logout
Enter your choice (1-7): 7
Logged out.
```

Tracking:

```
Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 3
Enter Donation ID: DON001

Donation DON001
---------------
Food : Biriyaani
Quantity : 20 Kg
Location : Chennai
Map : 13.0878, 80.2785 (Chennai, India)
Available Until : 8:00 PM
Contact : Paul, 98765 43210
Donor : DNR002 - Paul (Donor)
Status : Delivered
Volunteer : VOL103 - Andrews (Volunteer)

Status History
  25 Oct 2026 07:08 PM - Available (DNR002) Donation created
  25 Oct 2026 07:09 PM - Claimed (VOL103) Claimed by volunteer
  25 Oct 2026 07:09 PM - Picked Up (VOL103) Food collected from donor
  25 Oct 2026 07:11 PM - Delivered (VOL103) Food delivered

Main Menu
1. Register User
2. Login
3. Track Donation
4. View All Donations
5. View Users
6. Exit
Enter your choice (1-6): 6
All records are saved. Goodbye!
```